

Know Who You Are: Earning Roles from Situated Peer Judgments

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Abstract

Multi-agent systems routinely assign work before they know which collaborator is useful for the particular dependency at hand. Fixed roles, global reputation, and terminal rewards can rank agents, but they do not establish whether a concrete delivery helped the recipient who had to use it, whether a later repair belonged to the producer, or whether that evidence should change the next assignment. We formulate this gap as role formation from situated peer judgments. Our protocol binds a producer delivery to a recipient’s observable use, revision, or rejection, filters the result through an explicit responsibility gate, publishes attributable evidence, and consumes that evidence in an assignment made before the next task executes. A delayed update then credits only the future responsibility opportunity and records complete collaboration cost. We call the resulting candidate mechanism responsibility-aware role evidence (RARE); its novelty claim is the identifiable closed loop, rather than a particular off-the-shelf updater. We specify a TeamBench-derived artifact track and a local-peer selection track, matched-information baselines, root-level splits, cost accounting, online-latency and forgetting measures, and falsification tests. The resulting protocol makes its claims testable: empirical result cells are reserved for independently frozen real-API runs.

CCS Concepts

• **Computing methodologies** → **Multi-agent systems**; *Cooperation and coordination*.

Keywords

multi-agent systems, role formation, peer judgment, online adaptation, responsibility

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Internal pre-results manuscript. The current scientific release gate is closed. The tables below define the experiment before confirmation data are observed; numerical result cells are intentionally blank.

1 Introduction

How can collaborators earn future responsibilities from the judgments of agents who use their work? Consider two constructed queue-processing cases with the same rejection and failed team

outcome. In one, a producer violates the priority-module contract; in the other, the module is correct but the recipient implements its consumer incorrectly. The same feedback calls for different responsibility assignments. Observing only rejection and terminal failure cannot distinguish the cases; the recipient’s account of what it used and why it failed may provide the missing evidence.

This example isolates responsibility ambiguity: feedback about a collaboration need not identify the producer’s contribution. The recipient has direct evidence of the delivery’s use, but its judgment can also reflect its own integration difficulties. Learning from that judgment requires separating these sources and testing whether the resulting evidence predicts useful future assignments. A later outcome can inform subsequent decisions; it cannot retrospectively alter the information available to an already sealed assignment.

This paper studies a sharper question: *when can the agent that actually receives and uses a delivery provide reliable evidence about the producer’s future responsibility?* The question has a necessary sequence:

delivery → recipient use and judgment → producer attribution
→ future assignment → unseen quality and cost. (1)

A final score cannot by itself identify which producer caused a defect. An acceptance bit cannot distinguish a useful artifact from an easy integration. A recipient’s normal integration work cannot automatically become a producer penalty. A role posterior that never changes a later pre-execution assignment is not role formation.

The central idea is to make the recipient’s local observation a public, version-bound evidence object with a responsibility boundary. A producer is credited only when the delivery was observed, the recipient action is traceable to that delivery, and a contract check excludes recipient-owned work. Evidence is published without silently updating the policy. Before a later task starts, a legal owner reads the snapshot and the selector seals its assignment and propensity. Only after the later task has an independent outcome can a delayed credit update occur. This separates what a recipient saw from what the future responsibility actually produced, and makes the separation observable in the ledger.

We name this protocol *earning roles*. The name describes a measurable process: a candidate earns responsibility opportunities through attributable evidence that changes future assignment under a fixed information and cost contract. The candidate implementation is called responsibility-aware role evidence (RARE). RARE is a method candidate, not a claim that a particular representation, neural head, RLS update, or periodic refit is optimal. The scientific claim to test is that the complete judgment–attribution–assignment loop identifies useful role evidence beyond raw acceptance, terminal-only outcomes, and same-information contextual trust.

The paper makes four concrete contributions as a testable design.

- We define an event and information contract that separates producer quality, recipient action, downstream adoption,



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final outcome, and UNKNOWN while preserving the exact delivery, version, read cut, and propensity.

- We give a two-stage role-evidence protocol: publication is immutable and does not change the policy; a later assignment is sealed before execution; delayed selected-only credit can affect only the future opportunity it evaluates. These boundaries make noninterference and idempotent replay testable.
- We predeclare a candidate ArtifactRole track for responsibility diagnosis, a secondary PeerSelect mechanism track, matched baselines, root-level splits, and complete resource accounting to distinguish information value from closed-loop team benefit.
- We state the falsification boundary. If recipient evidence has no out-of-sample information, if a same-information trust baseline explains the effect, if attribution is unsafe, or if quality gains disappear at matched complete cost, the role-formation claim does not pass.

The current draft does not report a positive efficacy result. Existing development traces qualify parts of the protocol and expose scorer and ownership failures; they do not establish role learning. This separation lets the final paper promote only claims supported by an independently frozen experiment.

2 Related Work and Positioning

Partner selection studies show that agents can learn whom to interact with under social or game-theoretic feedback [4, 5]. Reputation mechanisms for language-agent teams similarly address cooperation collapse through shared assessments [3]. These works motivate adaptive partner choice, but the evidence object in our setting is narrower: a concrete versioned delivery is evaluated by the recipient that depended on it, and the resulting evidence must be attributable to a producer before it can affect a future responsibility opportunity.

Other multi-agent systems optimize graphs, debate roles, or workflow structure [2, 6–8]. Such systems can change routing or role prompts, yet a changed route does not identify whether a producer’s artifact caused a recipient’s action. Our comparison therefore treats these systems as conceptual neighbors and requires an explicit adapter before using any one as a scientific baseline. A central controller, a raw acceptance rule, and a contextual bandit are stronger alternative explanations than a static role prompt because they can consume comparable histories under a matched budget.

The distinction from ordinary online learning is temporal and semantic. A source judgment is available before a later assignment, but the later quality outcome is available only after that assignment has executed. Updating the source producer from the later outcome would leak future information and conflate source and target responsibility. Our protocol makes the two events explicit and reports the service cost of maintaining the separation. Bandit estimators provide useful update comparators [1]; they are not themselves the claimed contribution.

3 Problem Formulation

3.1 Objects and information boundary

At episode t , the task supplies context x_t and a legal candidate set C_t . The selector chooses one candidate before any candidate in C_t

executes:

$$a_t \sim \pi_\theta(\cdot \mid x_t, C_t, R_t), \quad a_t \in C_t, \quad (2)$$

where R_t is the public evidence snapshot available at read cut w_t . Only the selected candidate is charged for execution. The candidate emits a versioned delivery o_t with digest d_t ; a recipient j_t reads that delivery and takes action m_t . The recipient may accept, revise, reject, or be unable to complete the integration.

The smallest auditable event is

$$e_t = (x_t, C_t, a_t, o_t, j_t, J_t, m_t, Q_t, Y_t, \tau_t, c_t), \quad (3)$$

where J_t is the recipient judgment, Q_t is the producer-contract check, Y_t is an independent later or terminal outcome when the task defines one, τ_t is feedback-arrival metadata, and c_t is the complete cost record. The event also stores the candidate version, task/root identifier, artifact digest, feature-schema version, selection probability $p_t = \pi_\theta(a_t \mid x_t, C_t, R_t)$, read-cut digest, and evidence lineage.

The policy may consume only public events whose arrival is at or before its read cut. Hidden gold, a private scorer, a future target outcome, another policy’s state, and an unexecuted candidate cannot affect selection, retry, stopping, or update. A missing or non-attributable event is UNKNOWN, not a negative label. This boundary is the difference between a role signal and a convenient retrospective score.

3.2 Responsibility gate

For a source episode, define $G_t = 1$ only when all of the following hold: (i) the producer contract or a pre-registered source defect is uniquely bound to the producer; (ii) the recipient inspected o_t and its action is traceable to o_t ; (iii) recipient-owned integration paths are empty; (iv) J_t, m_t, Q_t , identifiers, versions, and digests are complete; and (v) the source event has not already been credited. Otherwise $G_t = 0$ and the source cannot create producer evidence.

This gate distinguishes four observations that are often collapsed: a producer-content defect, a recipient’s own integration work, a sink or final-task failure, and an unresolved attribution. The public record stores the reason for $G_t = 0$. This makes safety measurable: a method that refuses ambiguous evidence may have a higher UNKNOWN rate while producing fewer false producer penalties.

4 Candidate Mechanism: Earning Roles

4.1 Two-stage evidence and update

The candidate mechanism has two independent interfaces:

$$\text{publish}(e_t) \rightarrow \{\text{RoleEvidence}, \text{UNKNOWN}, \text{invalid}\}, \quad (4)$$

$$\text{choose}(x_t, C_t, R_t) \rightarrow (a_t, p_t, \delta_t), \quad (5)$$

$$\text{validate}(j_t) \rightarrow \{\text{credit}, \text{UNKNOWN}, \text{invalid}\}, \quad (6)$$

$$\text{update}(\text{credit}) \rightarrow \{\text{updated once}, \text{no-op}\}. \quad (7)$$

Publication may change the public input to a later assignment, but it does not call the persistent updater. For a later target assignment j , credit is legal only if the assignment was written before target selection and task start, the selected candidate and propensity match the sealed preview, the target delivery and outcome are complete, and the assignment has not already been credited. The update then evaluates that future responsibility opportunity, never the source episode.

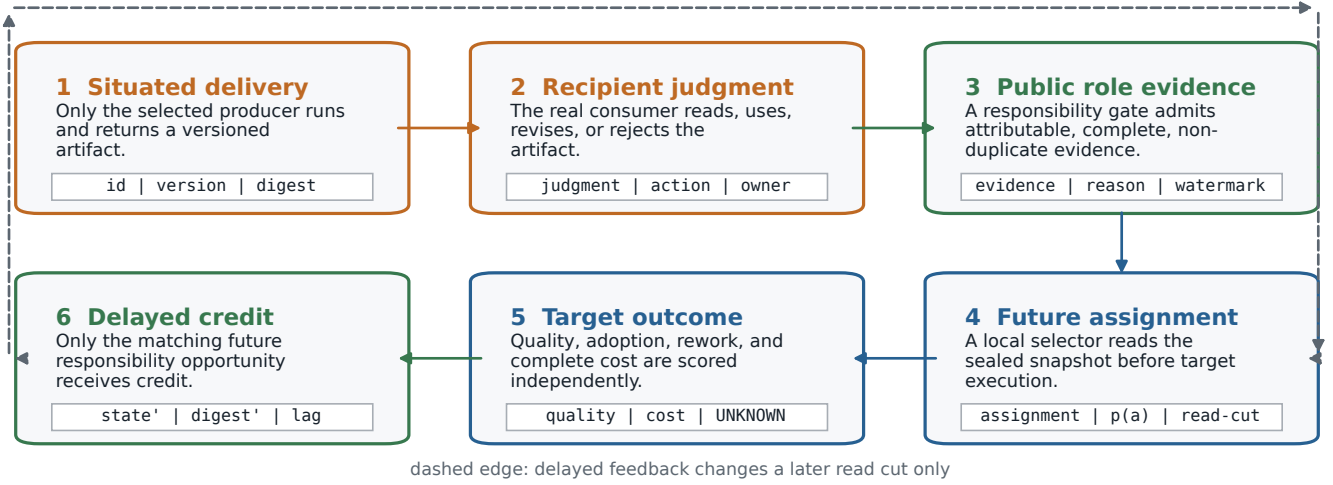


Figure 1: The proposed earning-roles loop. Orange marks situated observation, green marks responsibility-safe evidence, and blue marks the future decision and its independently scored consequence. The feedback edge is delayed: it can change a later read cut but cannot rewrite a sealed assignment. Each artifact shown in the diagram is a planned ledger field, not a reported result.

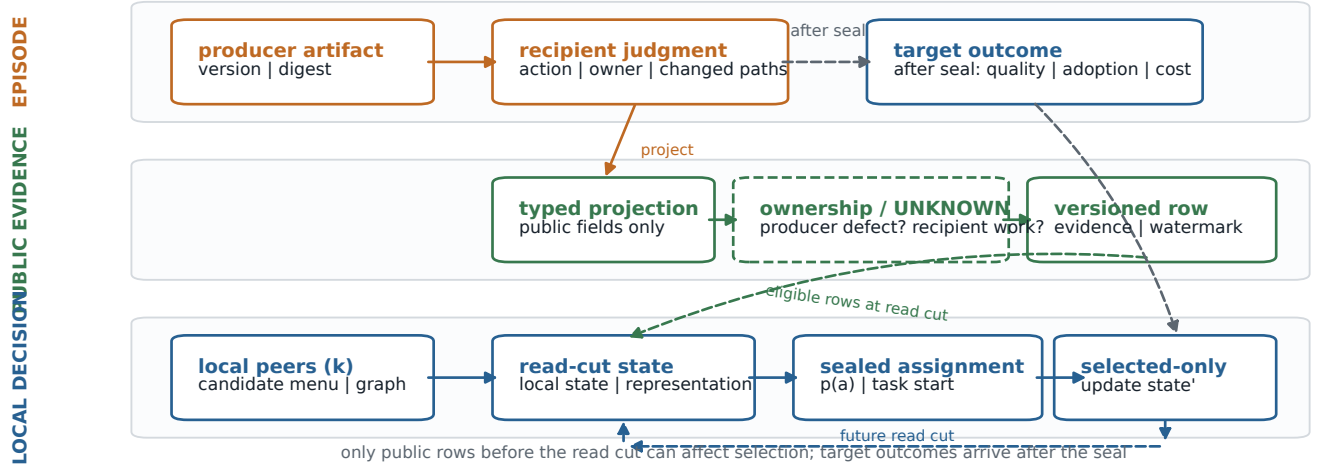


Figure 2: Evidence-to-role state. The episode lane records what the producer and recipient actually do; the public-evidence lane projects only typed, responsibility-checked fields; and the local-decision lane reads eligible rows at a read cut before sealing an assignment. Target outcomes arrive after the seal and update only the matching selected opportunity. The representation and updater remain replaceable interfaces.

The event-time contract is therefore

$$\begin{aligned} &\text{source selection} \rightarrow \text{delivery} \rightarrow (Q, J, m, Y) \rightarrow G \\ &\rightarrow \text{evidence publication} \rightarrow \text{assignment} \rightarrow \text{target outcome} \quad (8) \\ &\rightarrow \text{delayed selected-only update.} \end{aligned}$$

A correction creates a superseding evidence version and affects only future snapshots. Duplicate, late, out-of-order, resource-failed, or invalid events are idempotent no-ops or explicit UNKNOWN records.

Ledger replay must reproduce the same state digest and choice sequence.

4.2 Evidence representation and candidate score

A minimal public state for producer v in context r is

$$R(v, r) = (n(v, r), s(v, r), u(v, r)), \quad (9)$$

where n counts eligible evidence, s accumulates responsibility-safe positive/negative labels, and u records unresolved or rejected evidence for audit. A candidate score can combine a task representation $h(x, v, r)$ with the public evidence term:

$$\text{score}(v \mid x, r) = w^\top h(x, v, r) + \beta \frac{s(v, r)}{n(v, r) + \alpha}. \quad (10)$$

The representation h , updater, and constants are open experimental conditions. The formula is a contract-level candidate, not a theorem or a final backbone choice. The selector uses a pre-registered exploration floor, records p_t , and uses preview–assignment–commit so that writing a future assignment cannot resample the selected candidate.

The full mechanism is intended to have three properties. First, responsibility safety prevents a recipient’s normal work from becoming producer credit or blame. Second, temporal validity ensures that a judgment may affect a later assignment without reading its future outcome. Third, bounded online service makes each legal feedback event incrementally usable while the experiment reports update latency, state size, and complete cost. These are properties to measure, not assumptions that the formula automatically satisfies.

4.3 What counts as innovation

The claimed object is the closed loop in which situated judgment becomes attributable evidence, evidence changes a legal future assignment, and that assignment has an unseen quality–cost consequence. Existing representations, frozen encoders, RLS, online logistic regression, periodic refitting, caches, and prompt templates are implementation or comparison choices. The required tests are matched composition, leave-one-mechanism-out ablations, evidence-content mutation, and a same-information contextual controller. If these controls reproduce the complete result, the closed-loop mechanism has no identifiable advantage.

5 Benchmark and Baselines

5.1 Two complementary tracks

The primary track, **ArtifactRole**, is a TeamBench-derived candidate benchmark. It contains producer artifacts, recipient dependencies, use/rework/rejection actions, responsibility checks, later assignments, downstream adoption, and complete cost. Candidate roots are DIST1 queue-race (development diagnostic) and PIPE3 stream-processing (conditional confirmation); renamed seeds are not independent roots. Promotion requires material visibility, write permissions, scorer coverage, root independence, and clean replay.

The secondary **PeerSelect** track uses a pinned local graph interaction benchmark for selected-only payoff, local candidate sets, online updates, drift, and service cost. It can test the update mechanism efficiently, but it cannot establish situated artifact judgment or producer attribution. The two tracks share event and selector interfaces while keeping task, label, scorer, denominator, and claim separate.

5.2 Predeclared baselines

All primary arms receive the same candidate menu, initial opportunities, model/API/tool budget, exploration opportunity, feedback

arrival schedule, failure handling, and complete cost accounting. Each arm reads only its legal history.

RLS, online logistic/SGD, periodic refitting, and any candidate incremental updater are training comparators. They are not counted as separate selection policies and are not presented as the innovation merely because they are fast.

6 Experimental Questions and Matrix

6.1 Research questions and estimands

RQ1 (information value). Does an eligible situated judgment predict an independent producer-contract or later-use result beyond raw acceptance and terminal-only feedback? The primary estimands are out-of-sample log loss, Brier score, AUC when defined, calibration, and incremental predictive value on an independent root.

RQ2 (assignment causality). Does public evidence change an assignment made before target execution, including when the future owner did not produce the source judgment? We measure assignment-change probability, propensity-weighted future quality, adoption, rework, and complete cost. Evidence from producer B cannot be consumed as if it belonged to producer C .

RQ3 (online behavior). Does the update path satisfy the service contract under selected-only delayed feedback? We report publish, read/assignment, and update p50/p95 latency, service lag, backlog, state bytes, CPU/GPU/RAM, token/tool/API calls, and replay time.

RQ4 (stability and boundaries). How does the method behave under task or peer drift, delayed or out-of-order feedback, version replacement, recipient-only work, mixed ownership, scorer failure, and resource failure? We report old-root peak and mean forgetting, drift recovery window, UNKNOWN rate, correction consistency, and false producer attribution.

The primary unit of uncertainty is an independently reset stream. Episodes inside one stream are dependent. We use paired stream-level estimates, bootstrap or randomization intervals over streams, and pre-registered multiplicity control for the primary contrasts. Missing, invalid, or unknown events remain in the denominator and are never silently recoded as failures or successes.

6.2 Complete experiment matrix

Table 2 is the execution skeleton. The result columns are deliberately empty; the confirmation manifest must bind every row to a root, split, policy, seed, budget, scorer, and raw log before any result is filled.

6.3 Budget, statistics, and reporting

Every model request reserves attempted calls, prompt tokens, maximum completion tokens, tool time, and storage before dispatch; errors release only unused reservations and retain their observed cost. We charge producer execution, recipient integration, judgment service, communication, retries, replay, verification, and returned artifacts. Development/search cost is reported separately from deployment cost. A pooled controller receives the same total capacity, and arms that do not use a judgment still pay the declared comparison cost or use a shared judgment call whose output is masked.

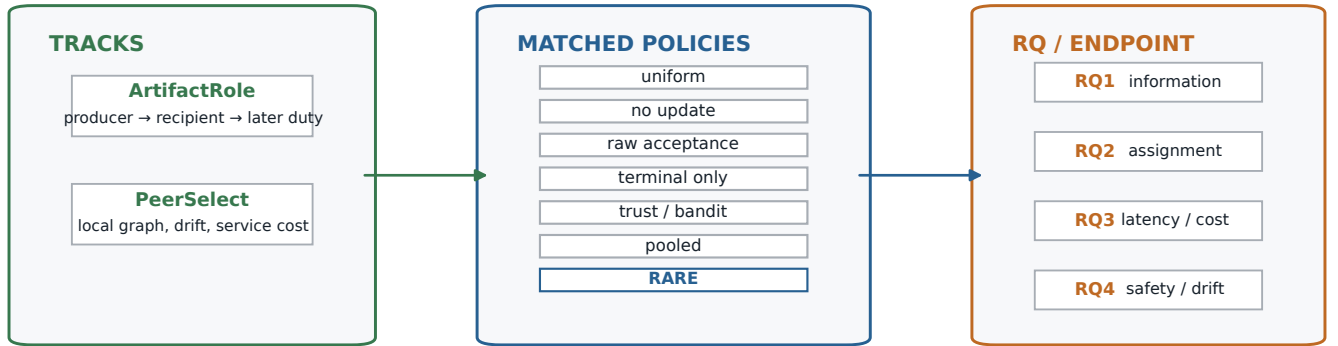
For each stream we record configuration, model route and version metadata, code and data hashes, seed, hardware, timestamps,

Arm	Read/update contract	Alternative explanation tested
Uniform	Candidate menu; fixed local-uniform sampling	Random allocation and exploration lower bound
No update	Initial prior and current context; feedback ignored	Static execution is sufficient
Raw acceptance	Recipient accept/reject only	Responsibility filtering adds no value
Terminal only	Independent final outcome only	Situated judgment adds no information
Contextual trust/bandit	Same context, judgment, propensity, delay, and budget as RARE	The effect is ordinary contextual trust or bandit learning
Pooled controller	Permitted public history under the same total capacity	A shared controller explains the gain
Closest published adapter	Faithful public-information adapter, if it passes qualification	Existing role/organization method is sufficient
RARE	Responsibility-safe evidence plus delayed selected-only credit	Candidate closed-loop mechanism

Table 1: Primary comparison boundary. A named arm enters the scientific matrix only after an executable adapter, information declaration, cost contract, and independent qualification pass.

RQ/cell	Track/root	Policy/intervention	Primary comparison	Measures	Result
RQ1-I	ArtifactRole / dev root	raw acceptance vs terminal only vs RARE	Same producer deliveries and recipient observations; legal feedback only	Brier/log loss, calibration, judge reliability	—
RQ1-II	ArtifactRole / confirmation root	contextual trust vs RARE	Same public read cut, delay, propensity, model and budget	Incremental prediction and cross-root transfer	—
RQ2-I	ArtifactRole / confirmation root	no evidence vs public evidence only	Preview-assignment before target execution	Assignment change, future quality, adoption, rework	—
RQ2-II	ArtifactRole / confirmation root	delayed update only vs public evidence + delayed update	Same target tasks and selected-only outcome	Future utility and complete cost	—
RQ2-III	ArtifactRole / confirmation root	RARE vs pooled controller	Same total public capacity and candidate menus	Quality-cost utility, concentration, propensity	—
RQ3-I	PeerSelect / dev streams	uniform, no-history, history-only, candidate updater	Local graph, selected-only payoff, fixed call/token budget	Regret/payoff, update p50/p95, state bytes	—
RQ3-II	PeerSelect / drift streams	updater with/without correction and replay	Pre-registered peer/task drift and delayed feedback	Recovery window, forgetting, backlog, replay equality	—
RQ4-I	ArtifactRole / all qualified roots	responsibility mutation matrix	Producer-only, recipient-only, mixed, unknown, resource failure	False attribution, UNKNOWN precision, no-op rate	—
RQ4-II	ArtifactRole / all qualified roots	leave-one-mechanism-out	Remove gate, public evidence, delay, locality, or updater	Main utility, safety, cost, interaction effect	—

Table 2: Pre-registered experiment matrix. Empty result cells are reserved for independently frozen real runs. A row is not a scientific result until its root, stream count, scorer, costs, and raw event manifest are complete.



every arm receives the same menu, lawful feedback schedule, opportunity, and complete-cost budget

Figure 3: Experiment map. ArtifactRole measures the situated producer–recipient phenomenon; PeerSelect isolates local selection, drift, and service cost. All policy arms receive the same candidate menus, lawful feedback schedule, task opportunities, and complete-cost budget before the four research questions are scored.

raw API responses, token usage, intermediate actions, failure traces, UNKNOWN reasons, and per-sample outcomes. We report means and medians with 95% stream-level intervals, effect sizes for pre-registered contrasts, per-root heterogeneity, ITT and per-protocol

denominators, and quality–cost Pareto points. The main endpoint is assignment-level future quality–complete-cost utility; source producer score is auxiliary and cannot replace it.

Table 3: Reserved headline results table. Values will be filled only from the frozen confirmation manifest; intervals are stream-level 95% intervals.

Endpoint	RARE	Best same-information arm	Difference / interval
Future quality	—	—	—
Complete cost	—	—	—
Quality–cost utility	—	—	—
Assignment change	—	—	—
Judgment Brier score	—	—	—
Update p95 / state bytes	—	—	—
Old-root forgetting	—	—	—
UNKNOWN / false attribution	—	—	—

6.4 Ablations and falsification

The minimum ablation grid is: (A) no evidence/no update; (B) public evidence only, with persistent state unchanged; (C) delayed update only, without new public evidence; and (D) public evidence plus delayed update. Additional interventions remove responsibility filtering, selected-only restriction, delay, locality, correction/replay, or the assignment binding one at a time while holding information, model, candidate menu, and budget fixed. A content mutation that changes a visible judgment while holding terminal quality fixed must change the comparator score if the comparator actually consumes that judgment.

The main claim is withdrawn if an evidence publication implicitly changes persistent policy; if a recipient repair is credited to a producer; if assignment or read-cut order is invalid; if unknown or duplicate events update state; if contextual trust explains the complete effect; or if gains disappear at matched complete cost. A null, a trade-off, or a qualified protocol result is scientifically valid, but it cannot be written as successful role formation.

7 Current Evidence and Expected Interpretation

Current real traces qualify engineering boundaries only. Two limited real-API chains logged source selection, recipient judgment, consumer execution, source snapshots, and a later assignment. Both consumers changed their own integration file, while the producer priority artifacts failed a separate import/constructor audit. The original four-check consumer score therefore did not cover the producer contract. The feedback mapping equaled its prior, so selection probabilities did not change. These observations motivated the responsibility gate, the separate producer audit, and the evidence-content mutation test.

The traces do not establish role specialization, predictive judgment, future quality improvement, or real-time training. The agents in those chains shared a fresh model and prompt without persistent individual experience, so identity labels cannot be interpreted as learned expertise. The benchmark and baseline matrix remain candidates until a second structural root, independent live histories, complete costs, and fair adapters qualify. No result cell in this manuscript is filled from those engineering receipts.

When confirmation evidence arrives, the interpretation follows a fixed truth table. Information gain without assignment or quality gain supports a predictive-signal claim only. Assignment gain without independent information leaves causal attribution unresolved. Quality gain with excess cost or unacceptable UNKNOWN rate is a trade-off, not a quality–cost win. A result at insufficient root or stream coverage remains exploratory. This prevents a posterior change or a single successful trajectory from being promoted to the paper’s central claim.

8 Discussion and Limitations

The mechanism is intentionally narrower than a claim that agents acquire general skills or that a workflow autonomously discovers experts. It studies responsibility opportunities in tasks with a real producer–recipient dependency. PeerSelect can isolate local online selection and drift, but it cannot replace ArtifactRole evidence. Conversely, an ArtifactRole result cannot be collapsed into a generic bandit payoff.

The main threats are judge error, selective exposure, root dependence, incomplete responsibility coverage, and cost inflation from recipient observation. The design measures judge reliability and selection propensity, preserves UNKNOWN records, and uses independent roots and streams. It cannot infer a universal role semantics from a single workflow family. If the public evidence is too sparse, the correct outcome may be a well-calibrated null or a narrower task-conditioned result.

The final backbone and updater remain open. They will be chosen only after the qualified tracks and strong same-information baselines expose a bottleneck. A frozen representation with a small incremental head, RLS or online logistic regression, periodic refitting, and a bounded candidate updater are comparison conditions. A large GPU run is justified only when a real update or representation bottleneck is measured and the experiment card fixes the challenger, budget, and confirmation split.

9 Conclusion

This paper defines a precise object for self-evolving multi-agent roles: a recipient’s situated judgment becomes responsibility-safe evidence, that evidence changes a future assignment before execution, and the later assignment is evaluated on unseen quality and complete cost. The protocol makes each arrow auditable and gives close alternatives the same information and budget. The current draft records the method contract, benchmark and baseline matrix, statistical plan, and falsification boundary while leaving empirical cells empty. The decisive result will be whether the complete loop improves future responsibility allocation beyond equally informed alternatives without sacrificing attribution safety, online service, stability, or cost.

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Table 4: Illustrative protocol trace. A confirmation run must record the visible fields and preserve the unavailable information; this table is not an empirical result.

Phase	Public record and legal effect	Information unavailable at this phase
Before selection	Candidate menu, public-row watermark, read-cut digest, and sealed propensity	Hidden gold, unexecuted-candidate outcomes, and any future target outcome
Responsibility	Judgment, action, owner, changed paths, ownership gate, and explicit UNKNOWN reason	A terminal team score used as a producer label
After seal	Target quality/cost and one selected-only update for the matching opportunity	A late or duplicate event without correction lineage

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